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MASTER ITEM CATALOGUING & AUTO PCODE ENGINE

Module 1: PCode Generation, Cataloguing & Master Item Engine

Comprehensive 10-week implementation roadmap, universal interchange matrix, legacy SKU cleansing (50,000+ items), dedicated engineering resource allocation, and milestone-governed payment structure for Popular Auto Spare & A/C Parts Co. W.L.L.

TOTAL MODULE FEE

BD 3,409.091

IMPLEMENTATION TIMELINE

10 Working Weeks

DELIVERY EFFORT

2.50 Engineering Months

MILESTONE TRANCHES

4 Verified Gates (25%)



50,000+

Legacy SKUs Cleansed



< 100ms

Search Response Latency



1-to-Many

OEM Interchange Cross-Ref



15-Day UAT

Verification Grace Period

Functional Scope & Technical Architecture Breakdown

Module 1 establishes the bedrock of the entire Popular ERP ecosystem by introducing deterministic, algorithmically generated unique Popular Codes (PCode) alongside a centralized item master with multi-brand hierarchy and OEM cross-referencing.

Smart PCode Generation Engine

Algorithmic generator producing unique alphanumeric tokens based on Category, Sub-Category, Make, Model, and Part Family, eliminating duplicate item creation across branches.

Multi-Brand & Vehicle Hierarchy

Structured relational taxonomy supporting Japanese, Korean, American, and European vehicle brands, model years, chassis codes, and engine specifications.

Universal Interchange Matrix

1-to-Many and Many-to-Many cross-referencing between Genuine OEM part numbers, Aftermarket brand codes (Denso, Bosch, Valeo), and Popular internal PCodes.

Fast Multi-Param Fuzzy Search

Full-text indexing with soundex and typo-tolerant search across Arabic/English part names, partial part numbers, OEM codes, and car models with <100ms response.

Dedicated Engineering Team & Resource Rate Matrix (BHD)

Pricing for Module 1 is determined directly by allocating the 5-person dedicated engineering team over 10 working weeks (2.50 calendar months):

Project Manager & Solutions Architect

BD 545.455

BD 136.364 / Wk • 2.50 Mo = BD 1,363.638

100% Dedicated

Back-End Lead Engineer (PHP MVC)

BD 227.273

BD 56.818 / Wk • 2.50 Mo = BD 568.183

100% Dedicated

Front-End Lead Engineer (React 18)

BD 227.273

BD 56.818 / Wk • 2.50 Mo = BD 568.183

100% Dedicated

Database & Cloud DevOps Specialist

BD 204.545

BD 51.136 / Wk • 2.50 Mo = BD 511.363

100% Dedicated

QA Automation & Multi-Branch Test Lead

BD 159.091

BD 39.773 / Wk • 2.50 Mo = BD 397.728

100% Dedicated

ENGINEERING DESIGNATION	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	2.50 Months (10 Wks)	BD 1,363.638
Back-End Lead Engineer (PHP MVC / REST APIs)	BD 227.273	BD 56.818	BD 1.420	2.50 Months (10 Wks)	BD 568.183
Front-End Lead Engineer (Admin UI / React Tokens)	BD 227.273	BD 56.818	BD 1.420	2.50 Months (10 Wks)	BD 568.183
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	2.50 Months (10 Wks)	BD 511.363
QA & System Test Automation Lead Engineer	BD 159.091	BD 39.773	BD 0.994	2.50 Months (10 Wks)	BD 397.728
TOTAL MODULE 1 DEDICATED ENGINEERING INVESTMENT				10 Working Weeks	BD 3,409.091

Milestone Delivery Breakdown & Payment Tranches

Module 1 is invoiced strictly across 4 equal milestone delivery gates upon verified demonstration and UAT staging sign-off:

Milestone Phase	Deliverable Description & Scope	Timeline	Share (%)	Invoice Amount (BHD)	Acceptance Verification Trigger
Milestone 1	PCode Database Schema, Legacy Data Cleansing Scripts & Base Token Generator	Weeks 1–3	25.00%	BD 852.273	Live demonstration of cleansed schema & algorithmic PCode generator
Milestone 2	Brand / Model / Spec Hierarchy & Automated Dynamic Category Code Engine	Weeks 4–5	25.00%	BD 852.273	Sign-off of Make/Model vehicle hierarchy and dynamic categorization
Milestone 3	Universal Cross-Reference / Interchange Matrix & 1D/2D Barcode Tagging	Weeks 6–8	25.00%	BD 852.273	Validation of OEM-to-Aftermarket interchange matrix and barcode output
Milestone 4	Fast Search Engine (<100ms), 50,000+ SKU Data Migration & Final Branch UAT	Weeks 9–10	25.00%	BD 852.273	Final production migration sign-off and multi-branch UAT acceptance
TOTAL MODULE 1 CONTRACT VALUE		10 Weeks	100.00%	BD 3,409.091	

Terms, SLA & Governance Protocols

1. 15-Day UAT Review SLA

- Popular Auto Spare is allocated a 15-calendar-day UAT review window per milestone to test all acceptance criteria.
- Any bugs discovered during UAT are remediated at zero additional cost within 48 business hours.

2. Source Code & IP Transfer

- Full proprietary ownership of PCode generation algorithms, schemas, and indexing routines transfers to Popular Auto Spare Co.
- Zero ongoing license fees or third-party proprietary royalties.

Stakeholder Authorization & Digital Sign-Off Console

By affixing digital signatures below, authorized executive representatives formalize the approval and scope ratification for **Module 1: PCode Generation & Item Master Engine**.

Popular Auto Spare Co. W.L.L

Client Executive Sign-off & Commercial Approval

No Signature on Record

PENDING SIGN-OFF

SaNDS Lab Middle East W.L.L

Lead Solutions Architect & Engineering Director

No Signature on Record

PENDING SIGN-OFF

UniGlobal Consultancy

Executive Strategic Advisor & Governance Lead

No Signature on Record

PENDING SIGN-OFF

